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Project Report

Fake News Detection System

1. Introduction

In the modern digital world, information is disseminated quickly via digital platforms. Yet, not all information is correct, resulting in the prevalence of the problem of fake news. The goal of this project is to create a machine learning-based fake news detection tool that categorizes the news as Real or Fake.

It uses Natural Language Processing techniques along with machine learning algorithms like Logistic Regression and Naive Bayes to analyze the data.

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2. Objectives

• To preprocess and clean textual news data

• To convert text into numerical features using TF-IDF

• To train machine learning models for classification

• To compare model performance

• To build an interactive system for user input prediction

• To save and reload trained models for future use

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3. Dataset

- The dataset is stored in a file named news.csv
- It contains the following columns:
    - **title**: Headline of the news ○

    text: Main content of the news

Data Preprocessing

- Missing values were removed
- Title and text were combined into a new column called **content**
- Text cleaning included: ○ Lowercasing

  ○ Removing URLs ○

  Removing HTML tags ○

  Removing punctuation ○

  Removing numbers

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4. Feature Engineering

4.1 Text Cleaning Function

A custom function was created using:

- Regular expressions
- String processing

4.2 TF-IDF Vectorization

- Converts text into numerical format
- Removes common English stopwords
- Ignores overly frequent words using max_df=0.7

4.3 Additional Feature

- **Text Length** was added to analyze distribution

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5. Data Labeling

Since the dataset was **unlabeled**, dummy labels were generated:

- 0 → Fake
- 1 → Real

Labels were assigned randomly using NumPy.

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6. Data Visualization

- Bar chart for label distribution
- Histogram for text length distribution

These visualizations helped understand:

- Balance of data
- Variation in content size

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7. Model Building

7.1 Train-Test Split

- 80% training data
- 20% testing data

7.2 Models Used

1. Logistic Regression

- A linear model used for classification
- Handles large feature spaces effectively

2. Naive Bayes (MultinomialNB)

- Based on probability theory
- Works well with text data

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8. Model Evaluation

Evaluation Metrics Used

- Accuracy Score
- Confusion Matrix
- Classification Report:
    - Precision
    - Recall
    - F1-score

Model Comparison

- Both models were evaluated
- The model with higher accuracy was selected as the **best model**

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9. Prediction System

A function predict_news() was created to:

1. Clean input text
2. Transform it using TF-IDF
3. Predict using the trained model

Output

- "FAKE "
- "REAL "

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10. User Interaction

- A loop allows users to enter custom news
- Typing exit stops the program

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11. Model Saving and Loading

Saving

- Model saved as model.pkl
- Vectorizer saved as vectorizer.pkl

Loading

- Models reloaded using pickle
- Verified using a test prediction

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12. Results

- Both models successfully classified news
- Logistic Regression or Naive Bayes selected based on accuracy

- System works interactively for real-time predictions

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13. Limitations

- Dataset was **randomly labeled**, so predictions are not realistic
- No real-world accuracy validation
- Limited feature engineering

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14. Challenges Faced & Learning Outcomes

- During the development of this project, several challenges were encountered, which contributed significantly to the learning experience.

- One of the major challenges was the **absence of labeled data**. Since the dataset did not include labels indicating whether news was real or fake, dummy labels had to be generated randomly. This created a limitation in terms of model reliability and real-world applicability. However, it helped in understanding the complete machine learning workflow.

- Another challenge was **text preprocessing**, as handling raw text data requires careful cleaning. Removing unnecessary elements like special characters, numbers, and URLs while preserving meaningful information required experimentation and fine-tuning of regular expressions.

- Feature extraction using TF-IDF also posed difficulties, particularly in understanding how to choose parameters such as stopwords and maximum document frequency. Incorrect parameter choices could lead to loss of important information or inclusion of irrelevant features.

- Model selection and evaluation were additional challenges. Comparing multiple models and interpreting evaluation metrics such as confusion matrix, precision, recall, and F1-score required a deeper understanding of classification performance.

- From a technical perspective, implementing **model saving and loading using pickle** was initially confusing, especially ensuring that the vectorizer and model remain consistent during prediction.

15. Approach to Solving the Problem & Key Decisions

- The approach to solving the Fake News Detection problem was structured in a systematic and step-by-step manner, following the standard machine learning pipeline. The first step involved understanding the problem of fake news classification and identifying suitable techniques from Natural Language Processing (NLP) and machine learning.
- The process began with data collection and preprocessing, where the dataset (news.csv) was loaded and cleaned. Since textual data often contains noise such as punctuation, URLs, and HTML tags, a custom text cleaning function was developed using regular expressions. A key decision at this stage was to combine the title and text columns into a single feature, as both contribute meaningful information for classification.
- Next, the cleaned text was converted into numerical form using TF-IDF vectorization. This decision was important because machine learning models cannot process raw text directly. TF-IDF was chosen over simpler methods like Bag of Words because it not only counts word frequency but also considers the importance of words across documents. Parameters like max_df=0.7 were selected to remove overly common words that do not contribute to classification.
- For model building, two algorithms were selected: Logistic Regression and Naive Bayes (MultinomialNB). The decision to use these models was based on their proven effectiveness in text classification tasks. Logistic Regression is efficient for high-dimensional data, while Naive Bayes is fast and performs well with probabilistic assumptions.
- Another key decision was to split the dataset into training (80%) and testing (20%), ensuring proper evaluation of model performance. Multiple evaluation metrics such as accuracy, precision, recall, and F1-score were used instead of relying solely on accuracy, providing a more comprehensive assessment.
- Additionally, a user interaction system was implemented using a loop, allowing real-time predictions. The decision to include model saving (model.pkl) and vectorizer saving (vectorizer.pkl) was made to ensure reusability and efficiency, avoiding retraining every time the program runs.

16. Learning Outcomes

This project provided valuable insights into both theoretical and practical aspects of machine learning and NLP:

- Gained a clear understanding of the end-to-end machine learning pipeline
- Learned how to preprocess and clean real-world textual data
- Understood the importance of feature engineering, especially TF-IDF
- Developed knowledge of classification algorithms like Logistic Regression and Naive Bayes
- Learned how to evaluate models using multiple performance metrics
- Gained experience in building an interactive prediction system

- Understood the importance of using real labeled data for meaningful results
- Improved problem-solving and debugging skills during implementation

17. Future Improvements

- Use a real labeled dataset
- Apply advanced NLP techniques (like Word2Vec, BERT)
- Deploy as a web application
- Improve accuracy with deep learning models
- Add more features like sentiment analysis

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18. Conclusion

This project shows the application of machine learning and NLP in the detection of fake news. Although dummy labels were used, the system correctly implements the entire process of:

- Data preprocessing
- Feature extraction
- Model training
- Evaluation
- Deployment (basic level)

This project can be a good starting point for the development of a real fake news detection system.

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19. References

- Scikit-learn Documentation
- Python Libraries: Pandas, NumPy, Matplotlib
- NLP Concepts (TF-IDF, Text Cleaning)

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